



Design Your Own Electric Vehicle Battery Charging Solutions

The application note below should help designers make their own electric vehicle battery charging solutions. If required, help is available from the company



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The popularity of electric vehicles (EVs) is increasing rapidly in India. According to a survey, the EV market in India is estimated to increase from 3 million units in 2019 to 29 million units by 2027 with a CAGR of 21.1 per cent. As a result, demand for AC/DC chargers, the smart chargers for EVs, will also increase.

In order to charge the batteries efficiently, and to ensure their long life, we need a smart battery management or charging system. To realise such an EV charging system, Holtek has come up with smart EV charger solutions based on their low-cost ASSP flash microcontroller (MCU) HT45F5Q-X for charging EV batteries.

At present, three EV charger designs suitable for Indian market—with specifications of 48V/4A, 48V/12A and 48V/15A—are available for rapid development of the

product. This semiconductor-based smart charging system can support both lithium-ion as well as lead-acid battery types.

Block diagram of the EV charging solution is shown in Fig. 1. Here, battery charger ASSP flash MCU HT45F5Q-X is the heart of EV charger circuitry, with in-built operational amplifiers (OPAs) and digital-to-analogue converters (DACs) that are necessary for battery charging function.

Specifications of the battery charger flash MCU HT45F5Q-X series are shown in Fig. 2. Designers can choose an appropriate MCU from HT45F5Q-X series according to their application requirement.

The features and working of EV charger solution for 48V/12A specification is briefly explained below. This EV charger design utilises HT45F5Q-2 MCU for implementing battery charging control function.

The MCU incorporates a battery charging

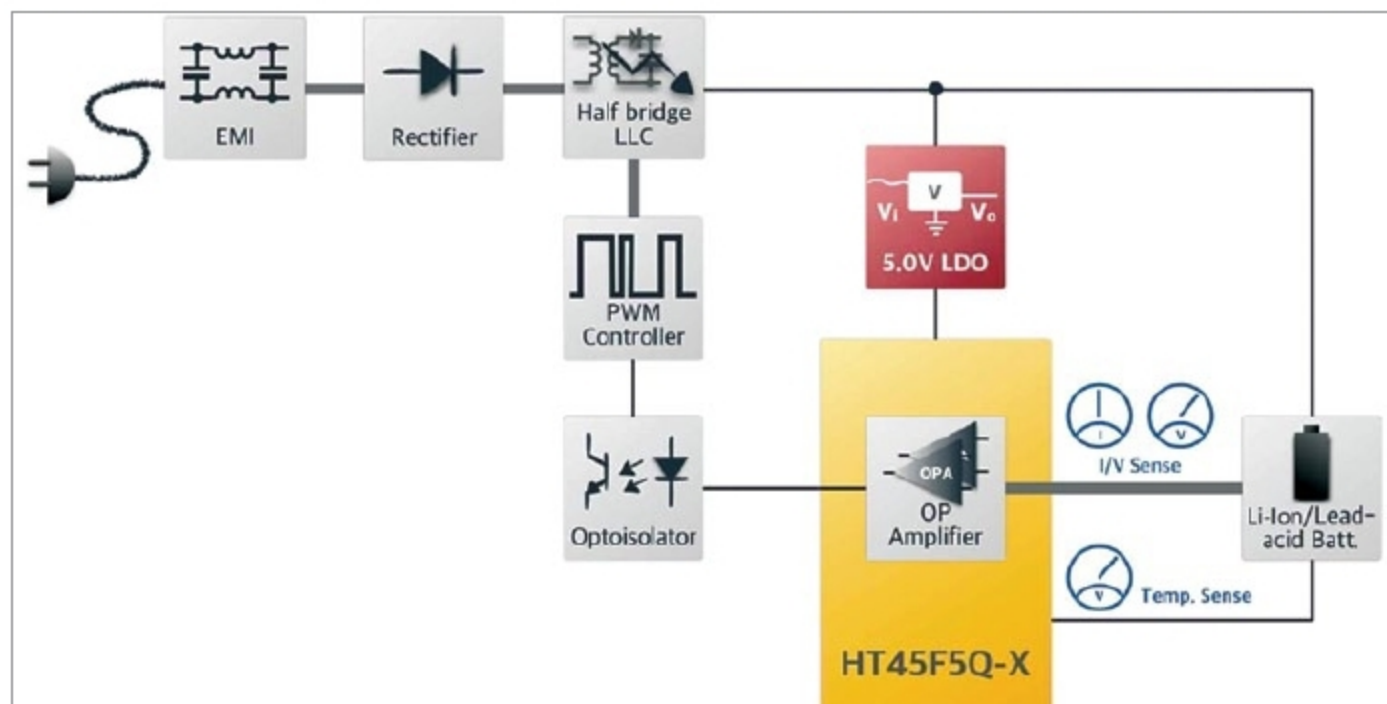


Fig. 1: EV charger block diagram